

Unusual topside ionospheric density response to the November 2003 superstorm

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Abstract

[1] We use observations from a variety of different ground- and space-based instruments, including ionosonde, ground- and space-based Global Positioning System (GPS) receivers, magnetometers, and solar wind data from the Advanced Composition Explorer (ACE), to examine the response of the ionospheric F2-layer height during the November 2003 superstorm. We found that the topside ionosphere responded unusually to the 20 November 2003 severe storm compared to behavior observed in a number of previous storms. While ground-based GPS receivers observed a large enhancement in dayside TEC, the low-Earth orbiting (~400 km) CHAMP satellite did not show any sign of dayside TEC enhancement. The real-time vertical density profiles, constructed from ground-based GPS TEC using a tomographic reconstruction technique, clearly revealed that the ionospheric F2-layer peak height had been depressed down to lower altitudes. Ionospheric F-layer peak height (hmF2) from the nearby ionosonde stations over Europe also showed that the dayside F2-layer peak height was below 350 km, which is below the orbiting height of CHAMP. The vertical $\mathbf{E} \times \mathbf{B}$ drift (estimated from ground-based magnetometer equatorial electrojet delta H) showed strong dayside downward drifts, which may be due to the ionospheric disturbance dynamo electric field produced by the large amount of energy dissipation into high-latitude regions. This storm demonstrates that data from LEO satellites varies widely among different superstorms.

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